THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE CHAPTER TEST SET A

Chapter 1: **Exploration: Entering the World of Chemistry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In science, a model is: 1

(a) an exact copy of the real object in every detail
(b) a simplified way of looking at a real system that keeps only what matters for the question
(c) a mathematical formula only
(d) a guess that has not been tested

2. In chemistry, atoms and molecules are commonly drawn as: 1

(a) spheres joined by bonds **(b)** flow charts
(c) layers of an onion **(d)** single points

3. An aircraft ran short of fuel because the ground crew mixed up: 1

(a) litres and gallons only **(b)** seconds and minutes
(c) metres and feet **(d)** pounds and kilograms

4. Newton's laws of motion explain the jerk felt when a bus stops suddenly. This is an example of: 1

(a) a unit conversion **(b)** a law describing regular behaviour
(c) an estimate **(d)** a guess

5. When a prediction does not match the observation, scientists: 1

(a) ignore the observation
(b) stop the study
(c) re-examine assumptions, models or measurements
(d) replace evidence with belief

6. The main purpose of making a rough estimate is to: 1

(a) replace all exact measurements **(b)** make science harder
(c) avoid using units **(d)** check whether an answer is reasonable

7. The word ‘work’ in science: 1

(a) means the same as effort
(b) is never defined
(c) has a specific meaning different from everyday use
(d) means any tiring activity

8. During the COVID-19 pandemic, designing good masks needed ideas from: 1

(a) physics, chemistry and biology together **(b)** only mathematics
(c) only chemistry **(d)** only art

9. ‘Scientific theory’ differs from the everyday word ‘theory’ because the scientific one: 1

(a) is the same as a belief
(b) is backed by careful testing and evidence
(c) is only an idea in someone's mind
(d) has never been tested

10. When studying how the heart pumps blood, a simple model can: 1

(a) assume the heart is a lamp
(b) include every atom
(c) ignore the blood
(d) ignore individual cells and treat the heart as a working system

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. What is a model in science? Give one example from physics. 2

12. Why is ignoring some details in a model not a mistake? 2

13. List two details that matter and two that can be ignored when modelling a cricket ball hit for a six. 2

14. Why does science use symbols and units agreed internationally? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Explain with an example how models are built and why assumptions are made. 3

16. Using the example of a cricket ball hit for a six, classify factors as important, minor or irrelevant. 3

17. A student wishes to model the time taken to cycle from school to home. Which details would she keep and which ignore? Why? 3

OR

Describe how the scientific meaning of the words ‘force’ and ‘work’ differs from everyday use. 3

18. State any three reasons why the symbol c is used for the speed of light and why such symbols are agreed internationally. 3

OR

Explain, using the aircraft fuel incident, why standard SI units are important. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A passenger aircraft ran out of fuel because the ground crew used the fuel density in pounds per litre instead of kilograms per litre. 5

(i) What type of error is this?
(ii) Why did the plane have to glide?
(iii) How do SI units help?
(iv) Name two quantities whose units must always be checked in such calculations.

OR

Case study: You want to predict whether a cricket ball will cross the boundary for a six. 5

(i) Which details are essential?
(ii) Which may be ignored in a simple model?
(iii) What extra details can be added later?
(iv) Why is this called model-building?

20. (i) Distinguish between a law, a theory and a principle. 5

(ii) Give one example each.
(iii) Why is a scientific theory not a ‘guess’?

OR

Case study: Varsha says it will rain this afternoon because the clouds are dark. Meghna asks questions. 5

(i) What makes a good scientific question here?
(ii) Write three such questions.
(iii) Why can forecasts still go wrong?
(iv) What does a failed prediction teach?

END OF QUESTION PAPER — **Total: 40 Marks**